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NMAP TO DISCOVER LIVE HOSTS USING ARP, ICMP AND TCP/UDP PING SCAN IN TRYHACKME

Aim: Learn & Implement Nmap host discovery techniques (ARP, ICMP variants, TCP, UDP pings) to identify live hosts on a network & compare their effectiveness

Description:
Use Nmap probe types to discover which IPs are active before port/vulnerability scans
Covers ARP, ICMP echo, TCP Syn, UDP Probes & combined scans

Definition:
Host discovery: Finding live systems
PR(ARP): Layer 2 MAC resolution
PE (ICMP Echo): Standard ping
PP (ICMP Timestamp) / -PM (ICMP (Mask)
PS (TCP syn): send Syn to specified ports
PA (TCP Ack): send ACK to specified ports
PU (UDP): send UDP to ports

Procedure:
ARP (LAN) nmap -sn -PR 10.10.X.0/24
nmap -sn -PR 192.168.1.0/24 -packet trace

ICMP Echo/Timestamp/Mask nmap -sn -PE 10.10.X.0/24 nmap -sn -PP 10.10.X.100
nmap -sn -pM 10.10.X.100

TCP syn/ACK (specify ports)
-PE 10.10.X.0/24 nmap -sn -PP 10.10.X.100
nmap -sn -PA 80, 443, 10.10.X.0/24

UDP nmap -sn -PU53,161 10.10.X.0/24
combined (disable ARP nmap -sn -PE
-PS -PA -PU 10.10.X.0/24 nmap -sn
10.10.X.0/24 --disable-arp-ping INPUT
Target IP or range (eg: 10.10.X.100 or
10.10.X.0/24)

Optional port lists for -PS, -PA, -PU
(eg: 22,80,443,53,161

Flags:- -sn, -PR, -PE, -PP, -PM, -PS,
-PA, -PU, - Packet-trace,
--disable-arp-ping

Output:
ARP
Nmap scan report for 10.10.X.1 Host
is up (0.00050s latency)
MAC address: 02:83:75:3A:F2:89
Nmap done: 256 IP addresses (3 hosts up)
ICMP
Nmap scan report for 10.10.X.100 Host
is up (0.00099s latency)
Nmap done: 1 IP address (1 host up)

Result: ARP discovery reliably identified hosts on local subnet with lowest latency & Highest Accuracy while ICMP, TCP (Syn/Ack) and UDP Probes found hosts